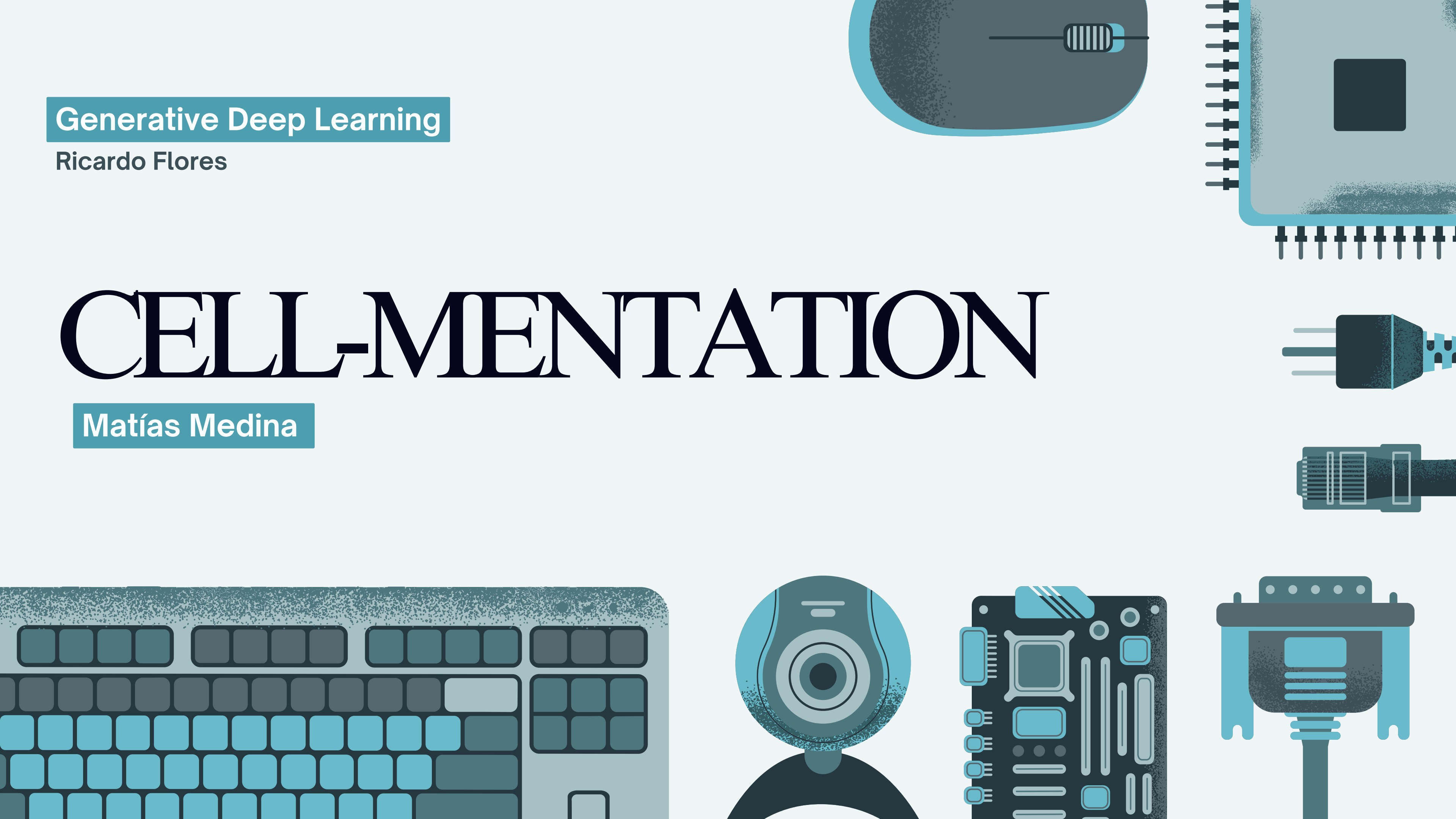


Generative Deep Learning

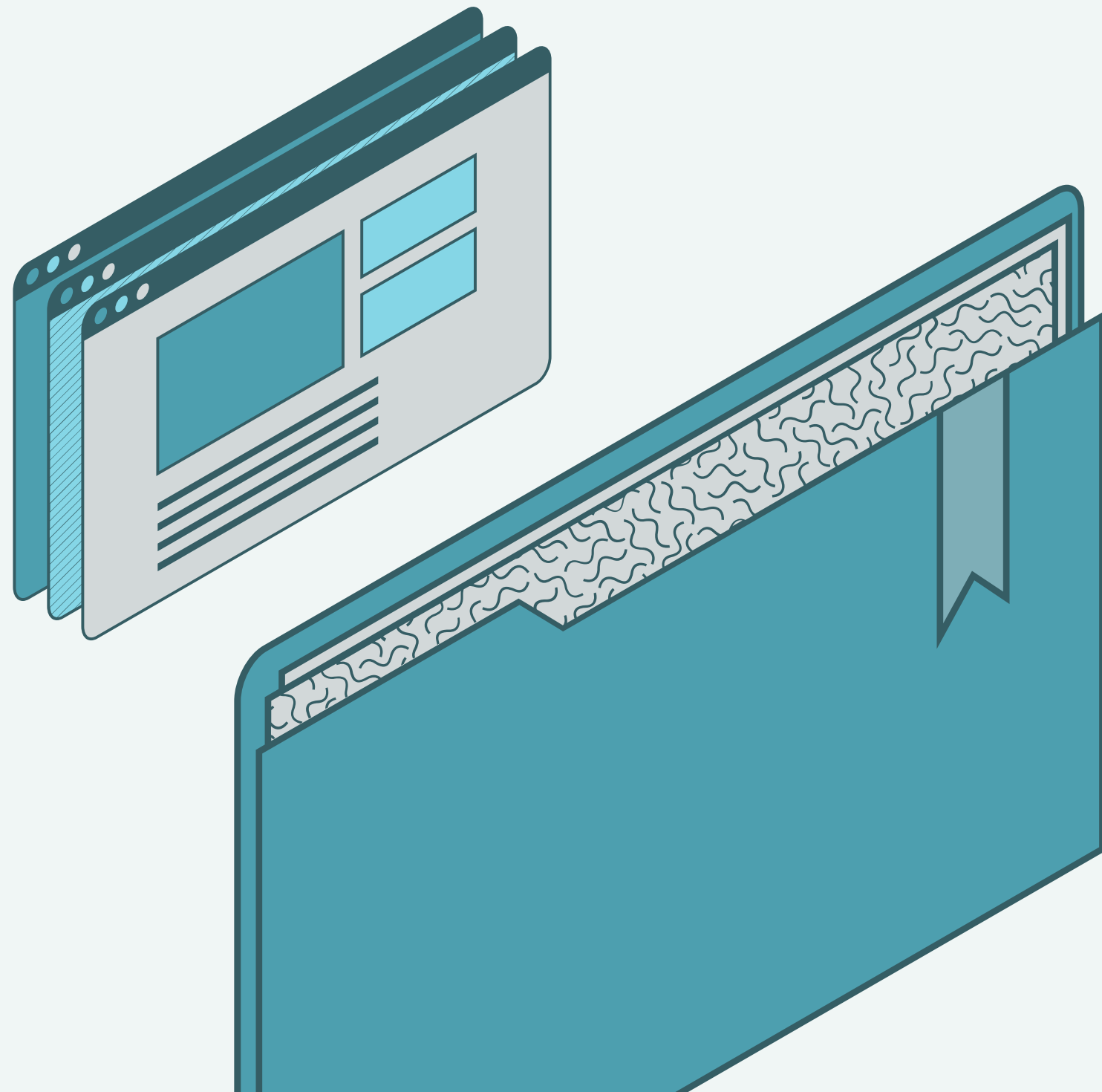
Ricardo Flores

CELL-MENTATION

Matías Medina



ÍNDICE



01. Introducción

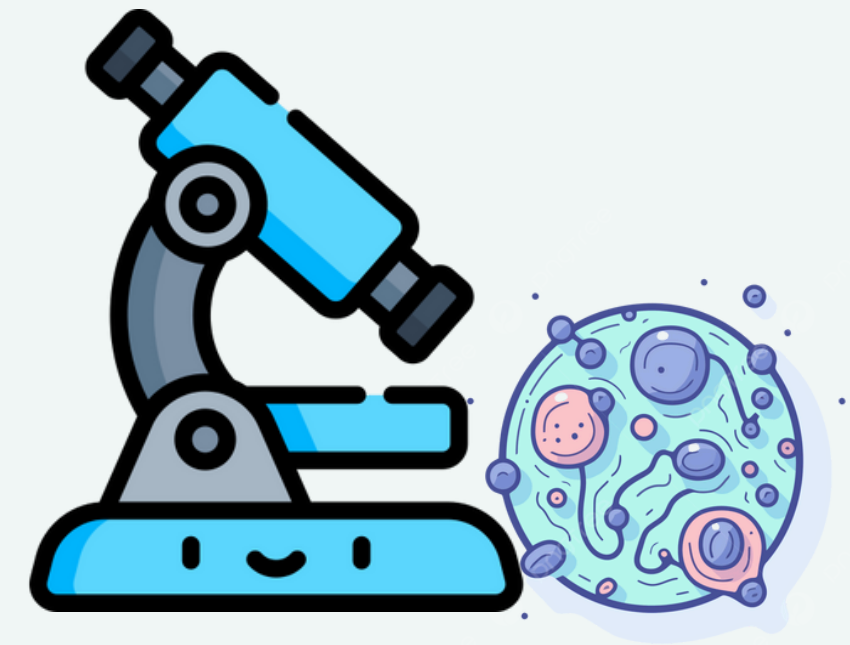
02. Dataset

03. Modelos

04. Resultados

Introducción

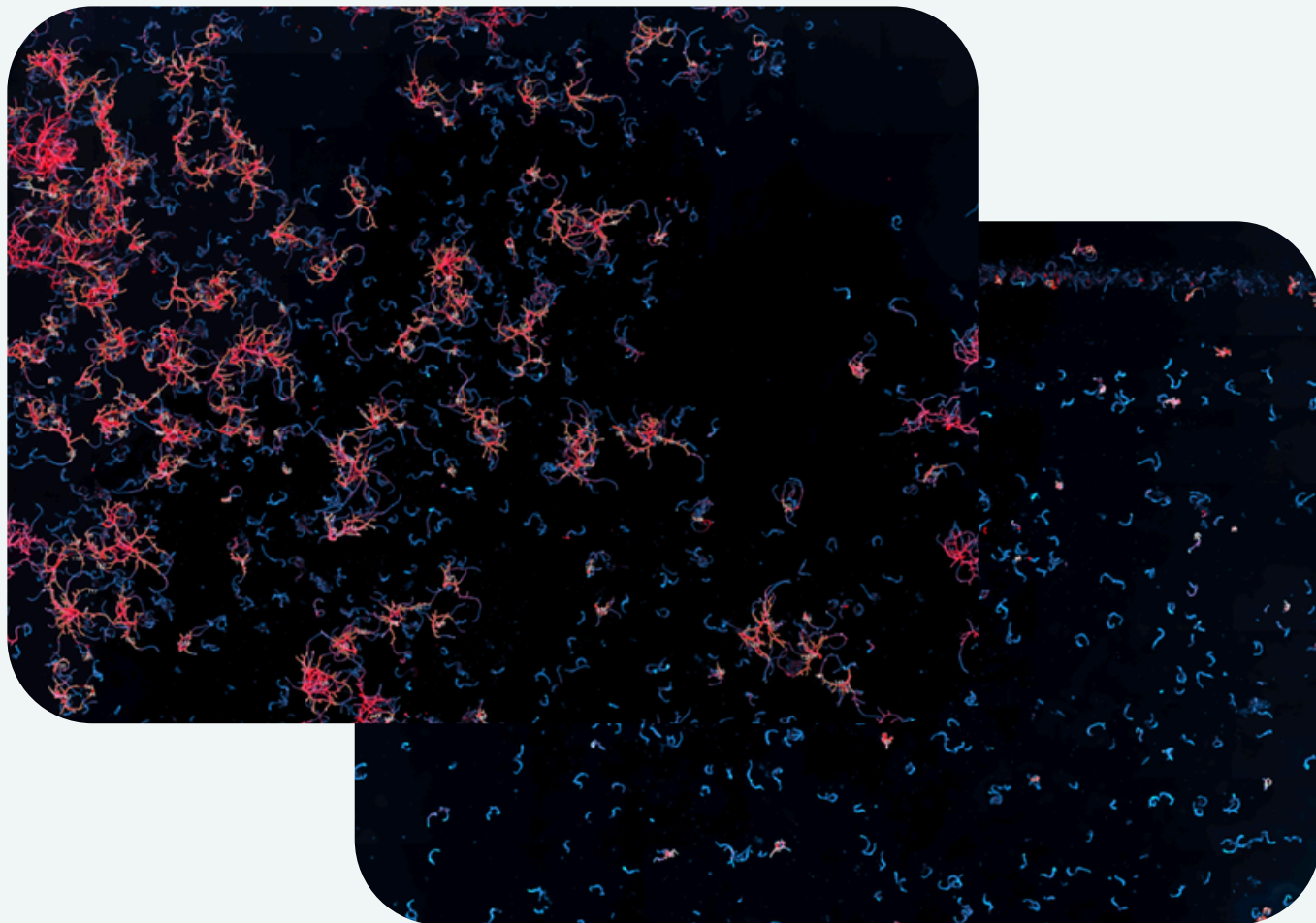
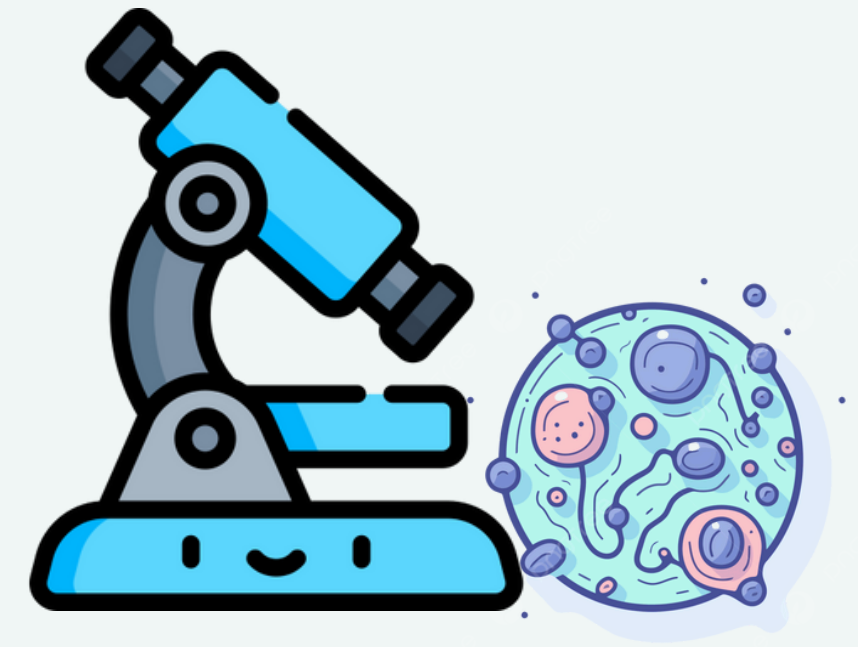
El problema se origina en el campo de la biología celular, donde los científicos estudian procesos fundamentales como la división celular, el crecimiento, etc.



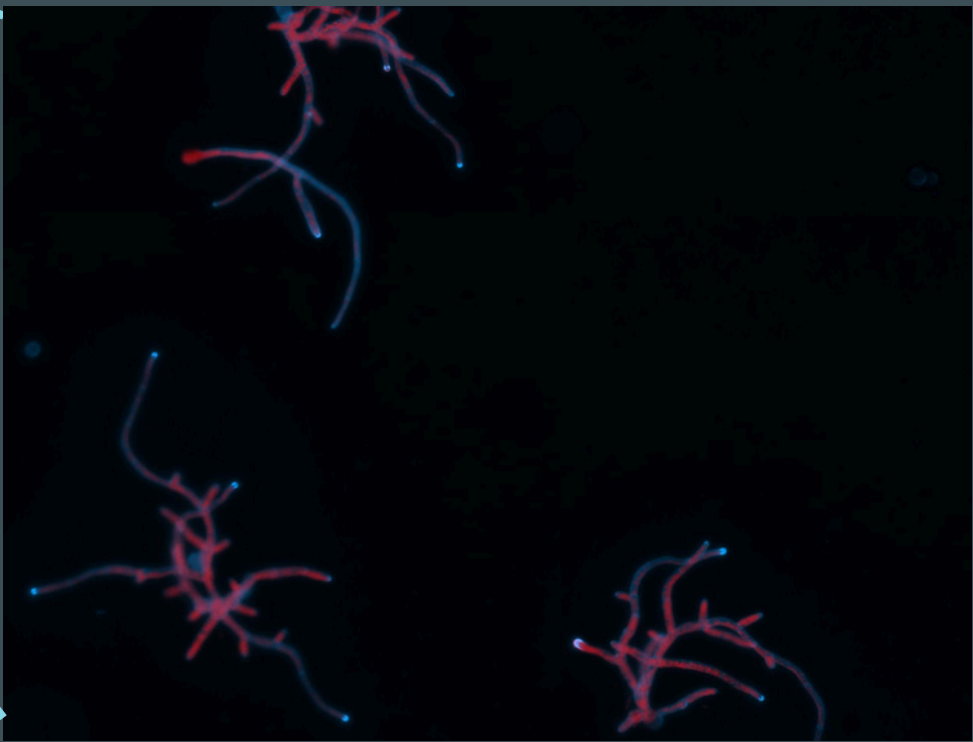
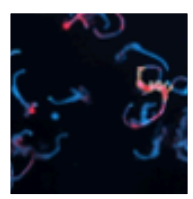
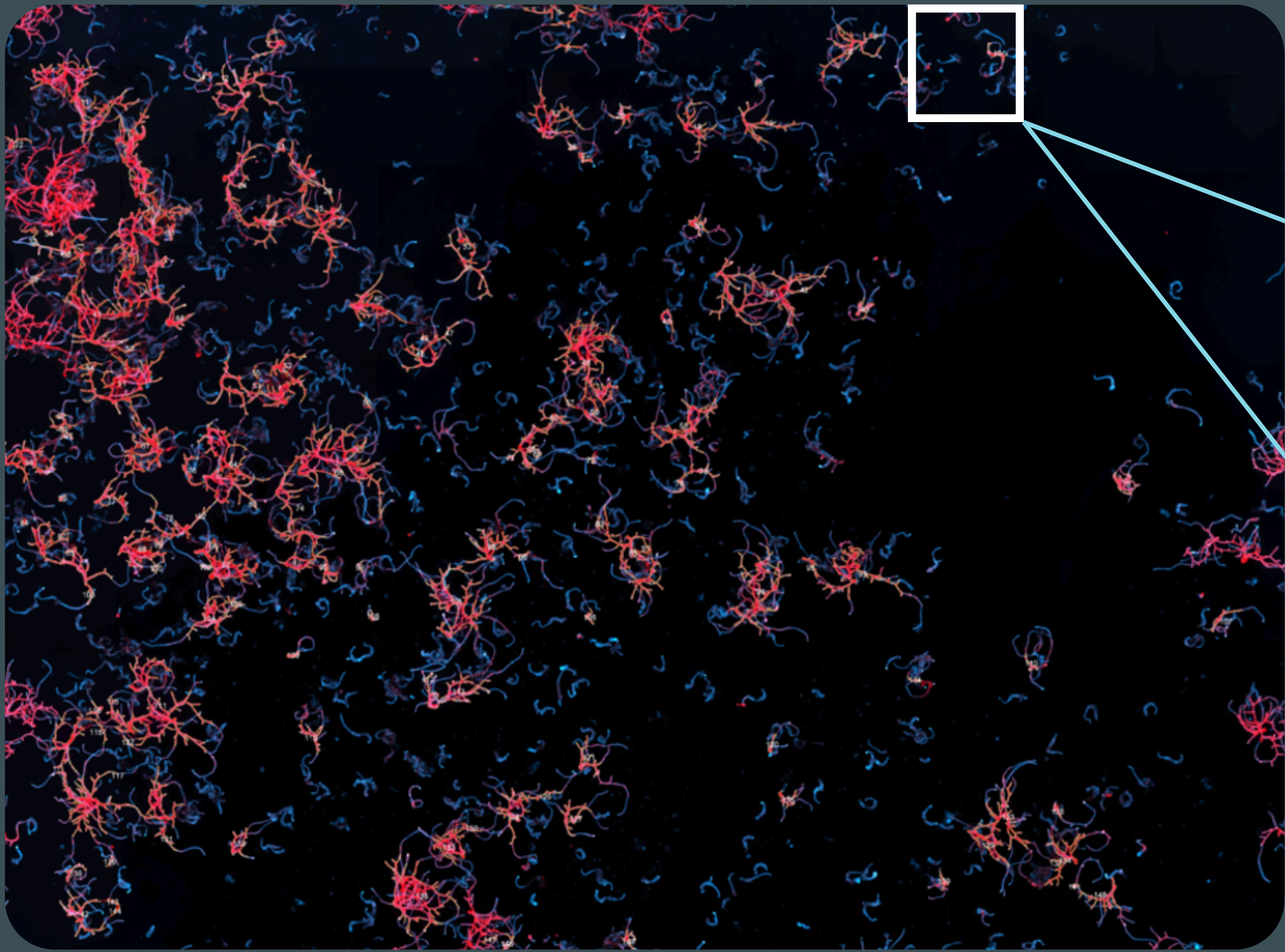
Se utilizan organismo de estudio el musgo *Physcomitrium patens*, debido a su alta capacidad de crecimiento y por su facilidad de manipulación.

Introducción

Uno de los objetivos de este tipo de estudios es medir la influencia que tienen distintos tratamientos en estas plantas.

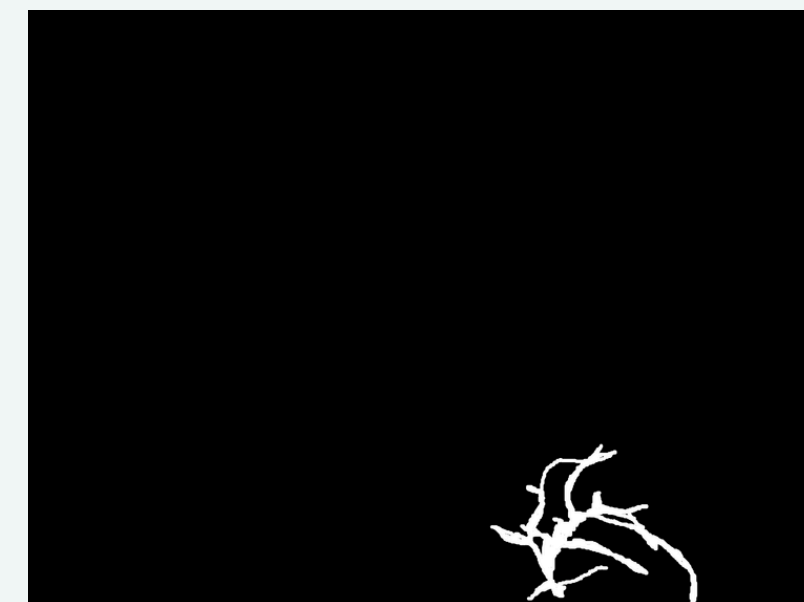
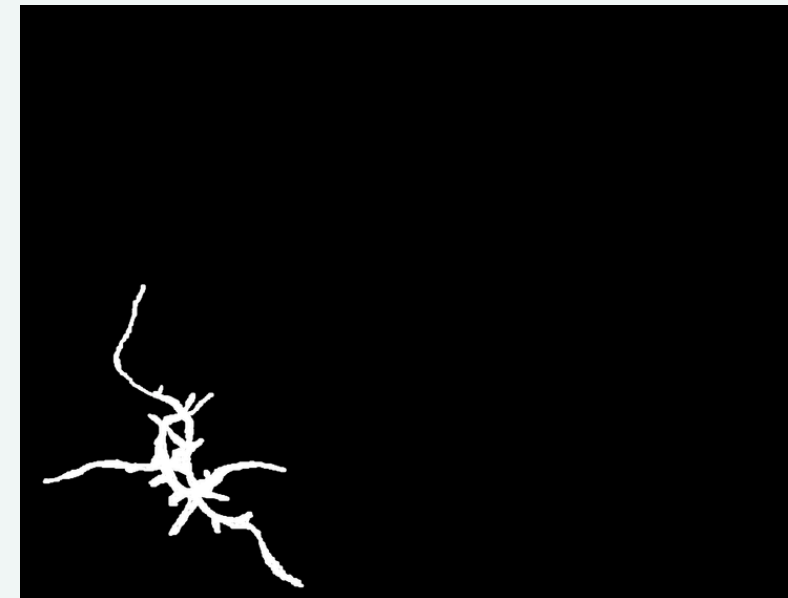
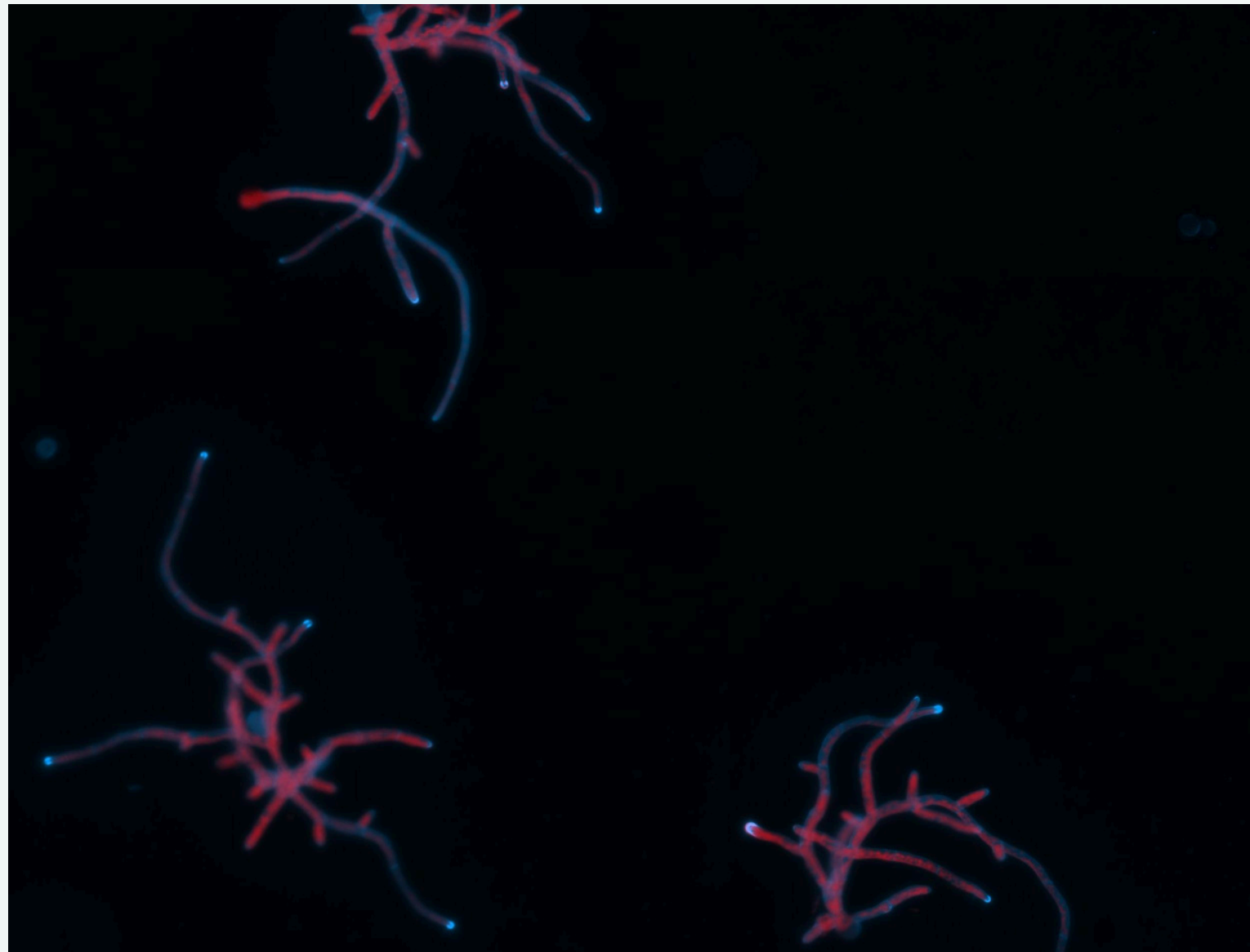


Mediante un microscopio automatizado, se capturan una enorme cantidad de imágenes en muy poco tiempo. Esto genera miles de imágenes que necesitan ser analizadas.



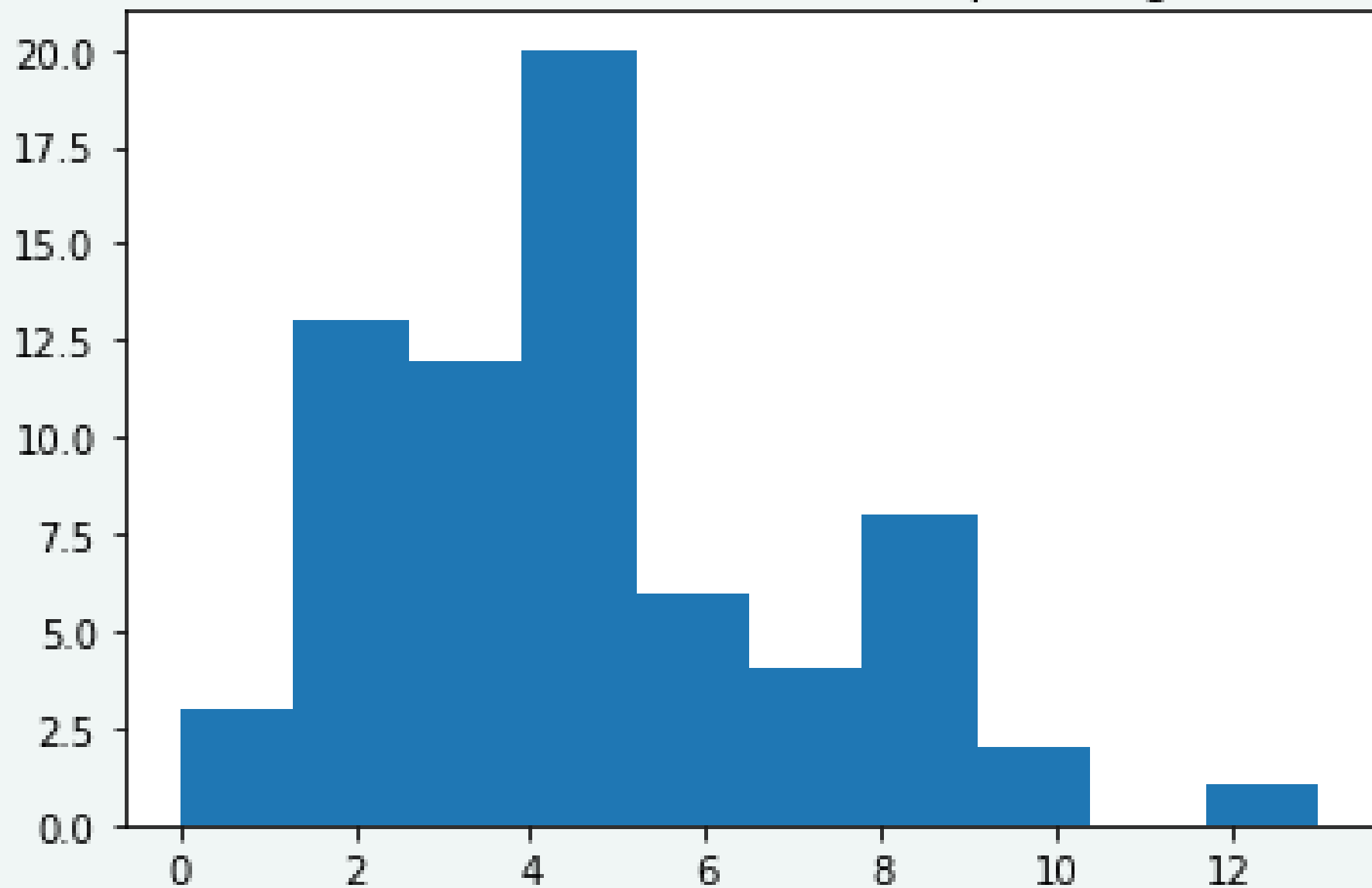
DATASET

- Las imágenes conseguidas mediante microscopia se cortaron en parches más pequeños para facilitar el entrenamiento del modelo a entrenar.
- Usando una herramienta llamada Label Studio, el equipo etiquetó manualmente las imágenes. En cada imagen, dibujaron sobre las plantas para enseñarle al futuro modelo exactamente qué píxeles corresponden a una planta viva, una planta muerta o ruido de fondo



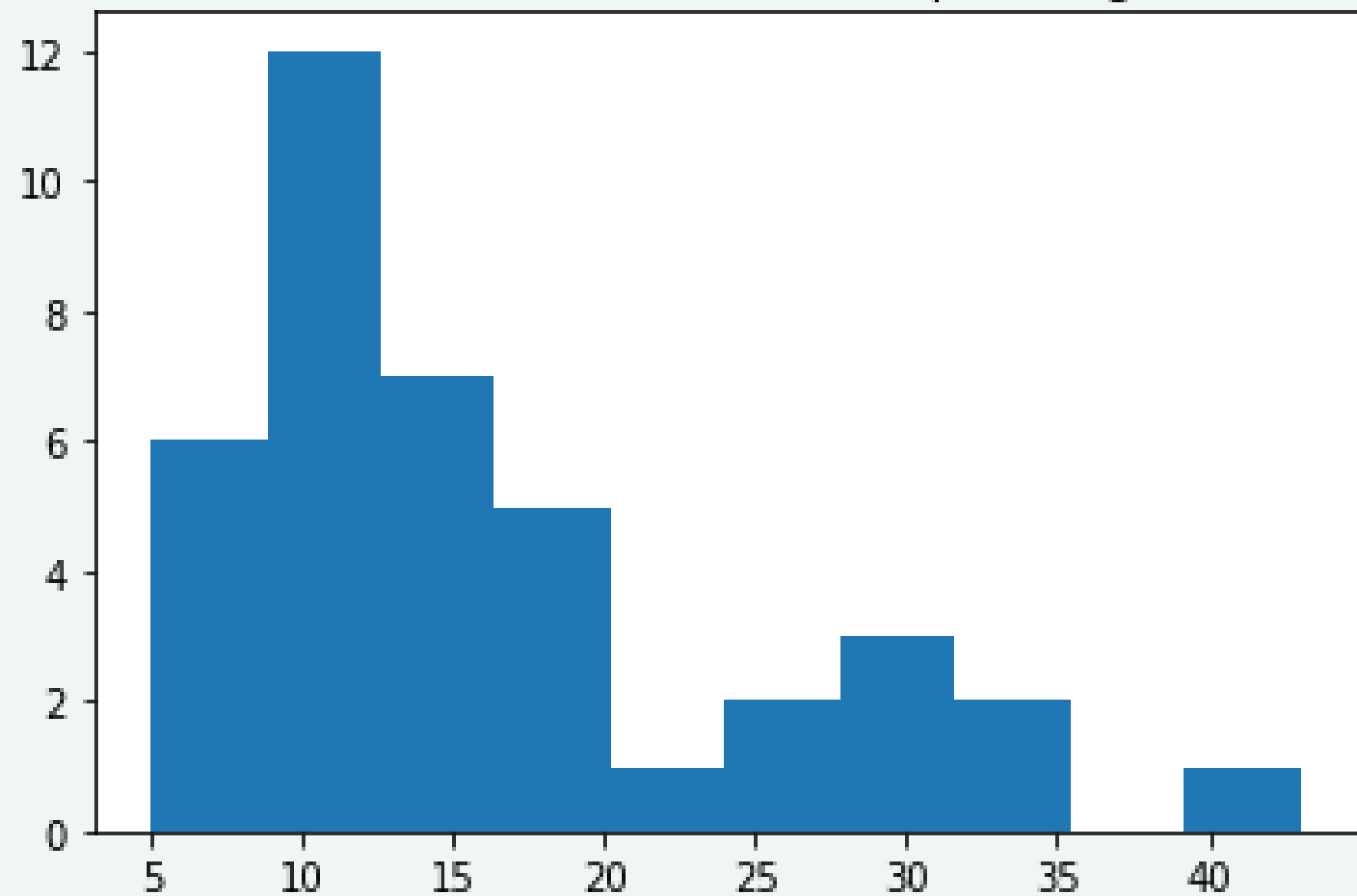
69 imágenes en “cwt”

Distribución del número de máscaras por imagen en cwt



39 imágenes en “dead”

Distribución del número de máscaras por imagen en dead



Segment Anything for Microscopy

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 Check for updates

Anwai Archit¹, Luca Freckmann¹, Sushmita Nair¹, Nabeel Khalid^{2,3}, Paul Hilt¹³, Vikas Rajashekar², Marei Freitag¹, Carolin Teuber¹, Melanie Spitzner⁴, Constanza Tapia Contreras⁴, Genevieve Buckley⁵, Sebastian von Haaren⁶, Sagnik Gupta¹, Marian Grade^{4,7}, Matthias Wirth^{4,7,8,9,10}, Günter Schneider^{4,7,11}, Andreas Dengel^{2,3}, Sheraz Ahmed² & Constantin Pape^{1,12}✉

CELLViT: VISION TRANSFORMERS FOR PRECISE CELL SEGMENTATION AND CLASSIFICATION

Fabian Hörst^{1,2*}, Moritz Rempe^{1,2}, Lukas Heine^{1,2}, Constantin Seibold^{1,3}, Julius Keyl^{1,4}, Giulia Baldini^{1,5}, Selma Ugurel^{6,7}, Jens Siveke^{8,9,10,11}, Barbara Grünwald^{12,13}, Jan Egger^{1,2}, and Jens Kleesiek^{1,2,7,14}

SwinNet: Swin Transformer Drives Edge-Aware RGB-D and RGB-T Salient Object Detection

Zhengyi Liu*, Yacheng Tan, Qian He and Yun Xiao

NUCLEI INSTANCE SEGMENTATION AND CLASSIFICATION IN HISTOPATHOLOGY IMAGES WITH STARDIST

Martin Weigert¹, Uwe Schmidt²

¹Institute of Bioengineering, School of Life Sciences, EPFL, Switzerland

²Independent Researcher, Dresden, Germany

MODELOS

SEGPLANTFORMER

- SegFormer es un modelo de segmentación semántica desarrollado por NVIDIA. Se basa en la arquitectura Vision Transformer (ViT).
- Se utiliza "nvidia/mit-b0" en el modelo original y "nvidia/mit-b2". Esto significa que no se empieza desde cero. Cargamos un modelo que ya ha sido entrenado en un dataset masivo (como ImageNet). Los parámetros ya tienen un conocimiento básico del mundo visual. Solo se realiza fine-tuning para la tarea específica de segmentar plantas.

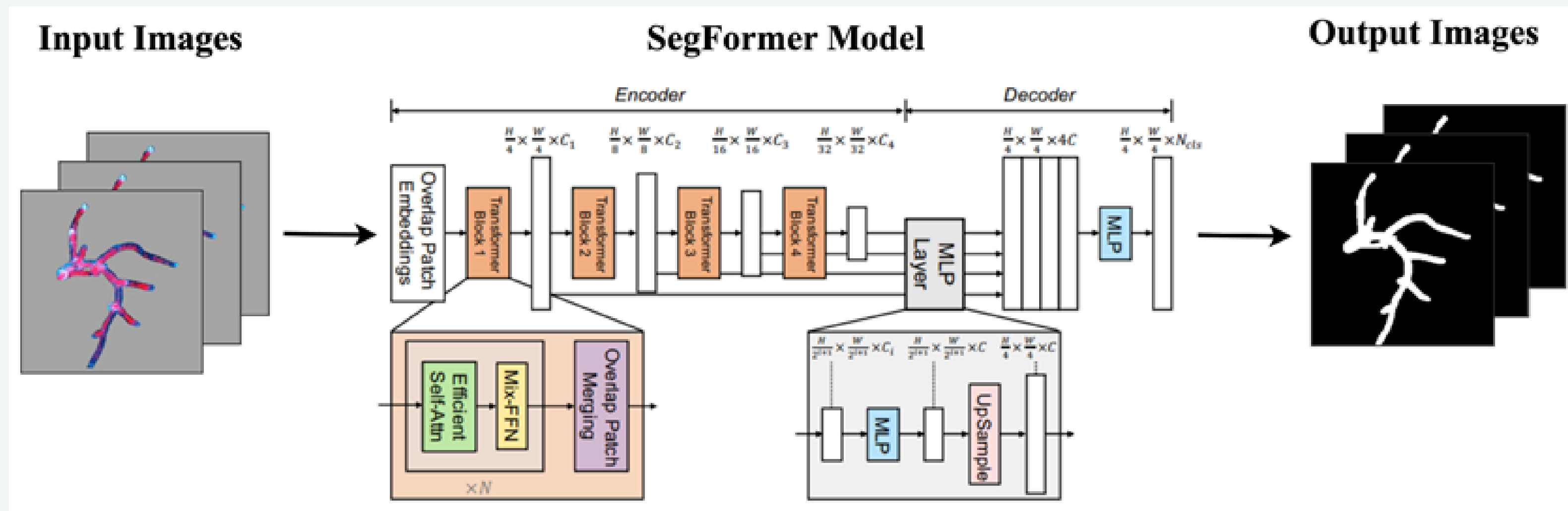


Image source: SegPlantFormer: A Transformer-Based Model for Plant Semantic Segmentation

MODELOS

SEGFORMER

IMG_SIZE = 512X512

BATCH_SIZE = 4

LEARNING_RATE = 1E-4

EPOCHS = 70

LOSS FUNCTION =
CROSSENTROPYLOSS

OPTIMIZER = ADAMW

MODELOS

U-Net

- Lo que hace a la U-Net tan poderosa es el “skip connections”. Son puentes que conectan directamente las capas del encoder con sus capas espejo en el Decoder. Esto permite que la información espacial de alta resolución que se pierde durante la compresión del encoder se reinyecte directamente en el Decoder.
- Backbone ResNet34:
 - Se implemento aprovechando la librería “segmentation-models-pytorch”
 - `unet_model = smp.Unet( encoder_name="resnet34", encoder_weights="imagenet", in_channels=3, classes=NUM_CLASSES, )`
- Backbone Transformer:
 - Se vuelve a utilizar el backbone segformer “MiT-B2”

MODELOS

U-Net

IMG_SIZE = 512X512

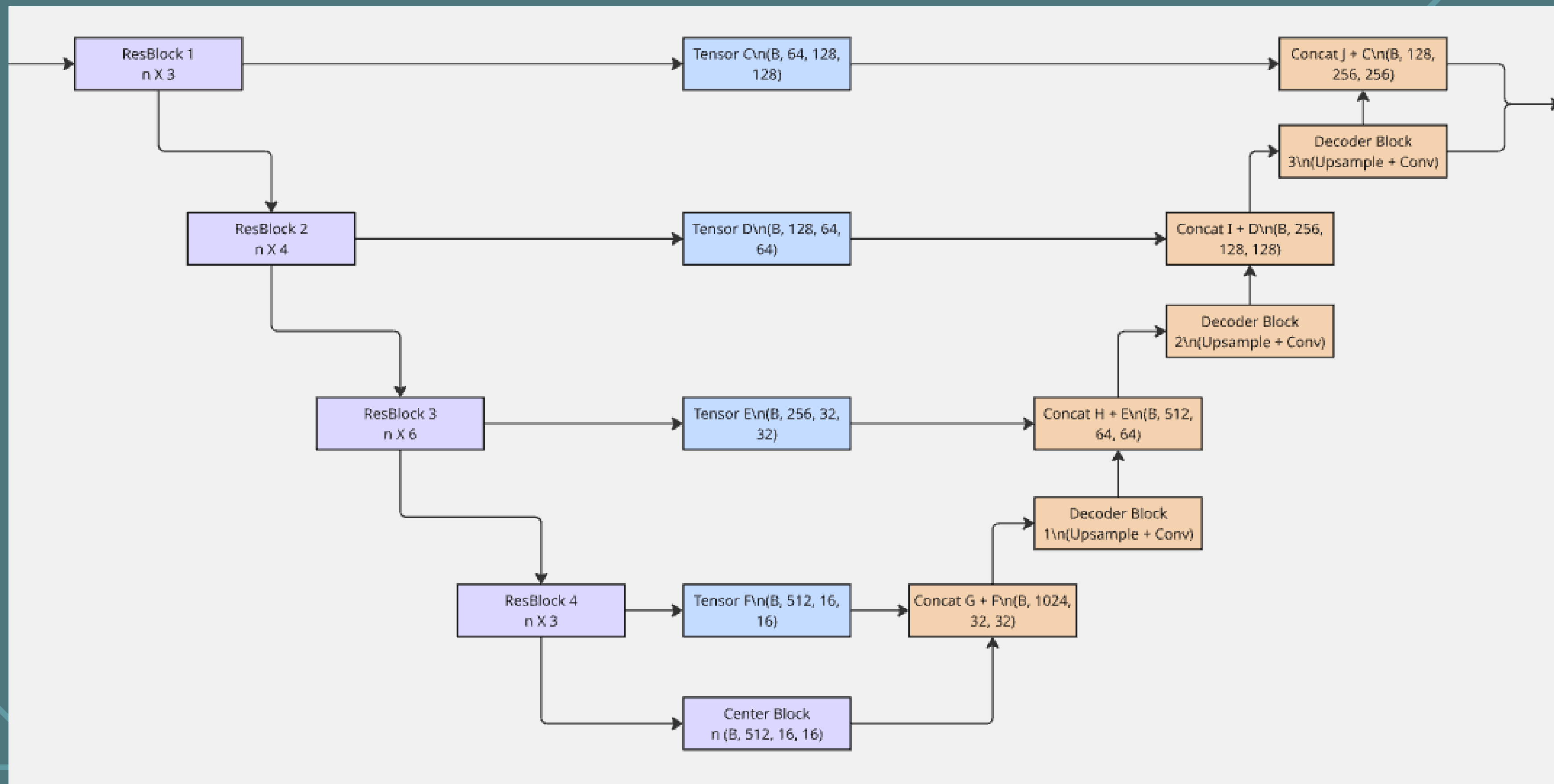
BATCH_SIZE = 4

LEARNING_RATE = 1E-4

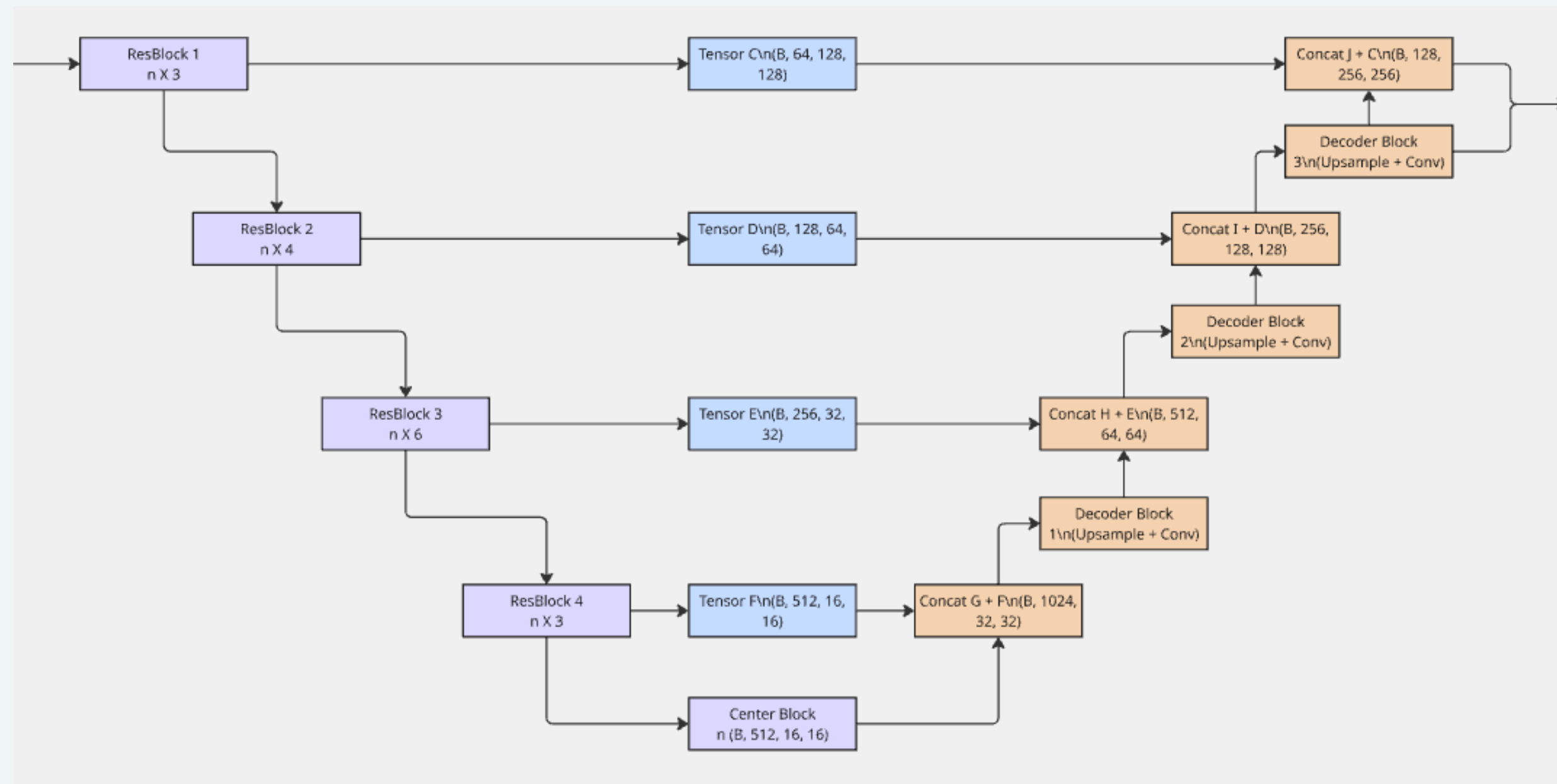
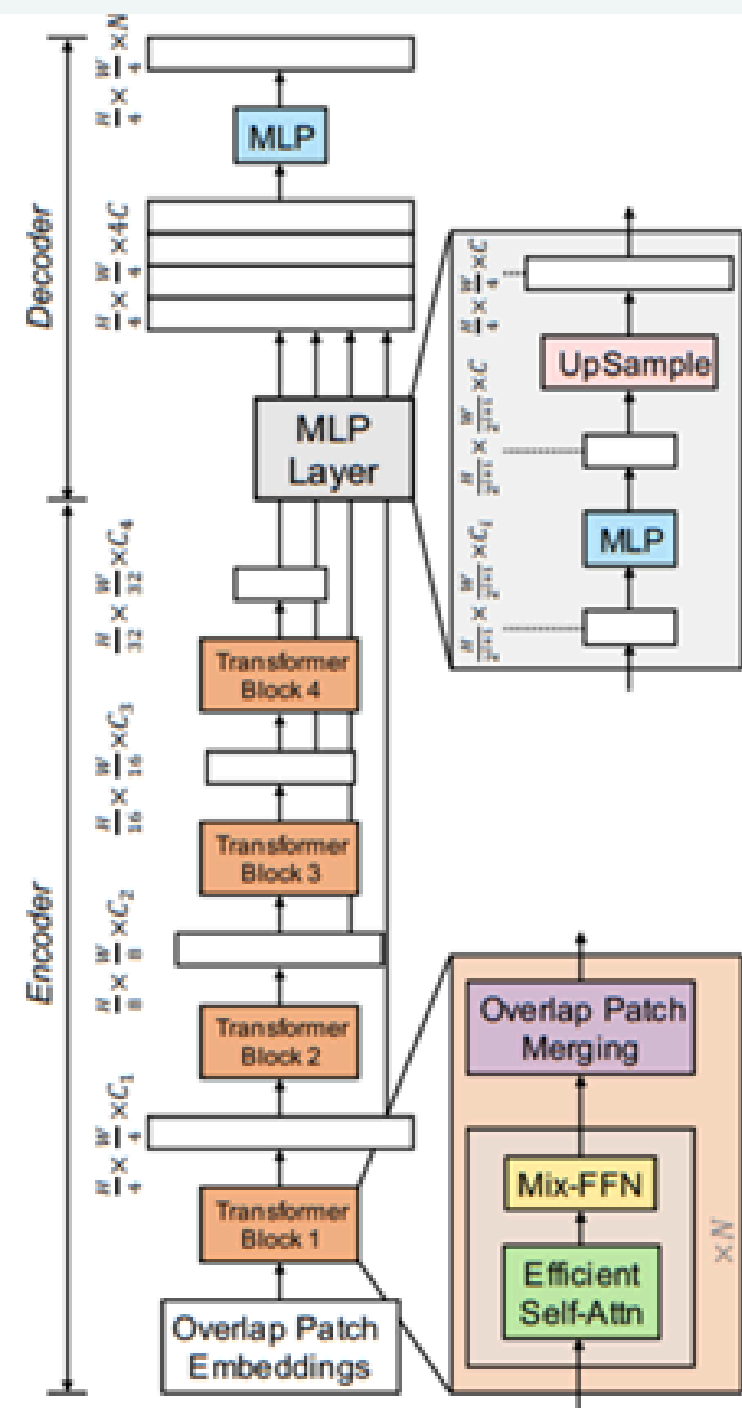
EPOCHS = 70

**LOSS FUNCTION =
CROSSENTROPYLOSS**

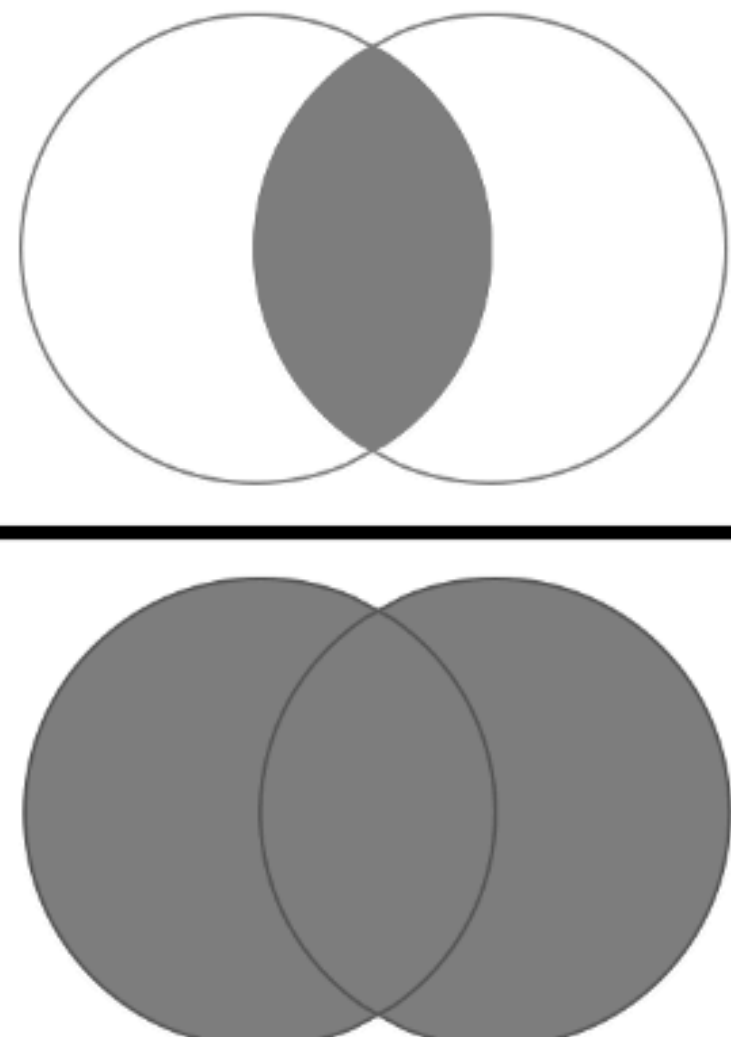
OPTIMIZER = ADAMW



SegFormer Model



mIoU

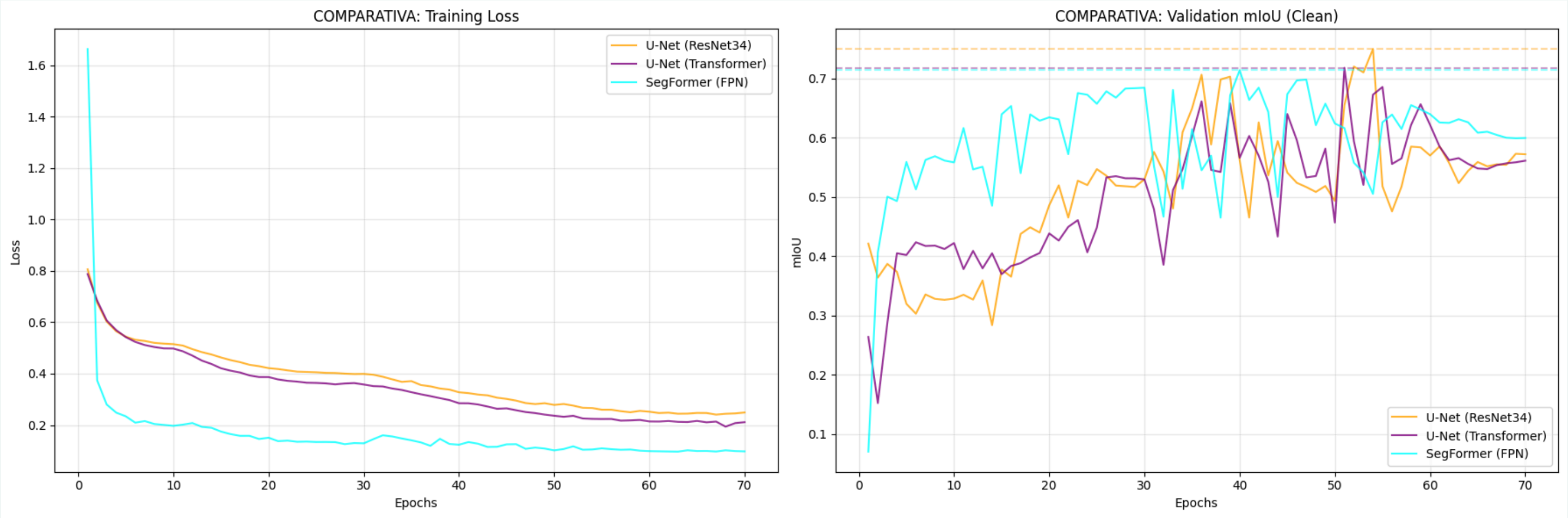
$$\text{IoU} = \frac{\text{Area of Overlap}}{\text{Area of Union}}$$


The diagram illustrates the Intersection over Union (IoU) metric. It consists of two parts. The top part shows two overlapping circles. The intersection of the two circles is shaded dark gray, representing the 'Area of Overlap'. The bottom part shows two overlapping circles. The entire area covered by both circles (the union) is shaded dark gray, representing the 'Area of Union'.

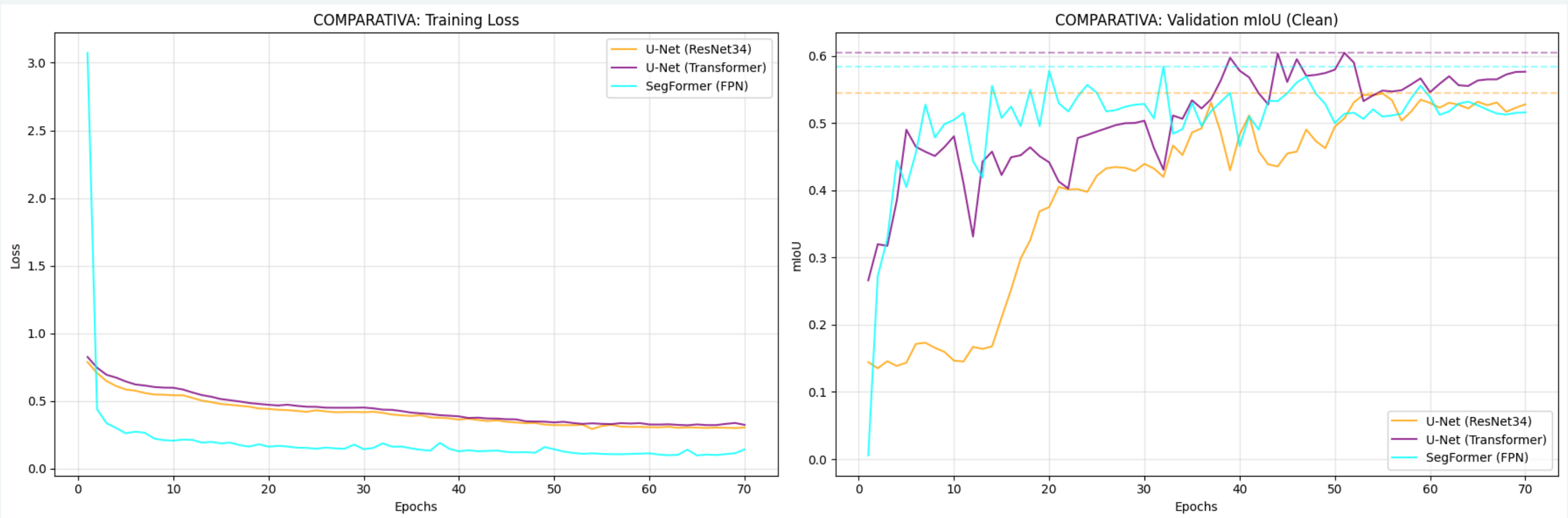
Resultados

Method	Backbone	CWT (Alive) mIoU	Dead (Treated) mIoU
<i>Traditional Methods</i>			
Baseline (ImageJ)	N/A	0.421	0.256
Human Expert	N/A	0.491	0.404
<i>Deep Learning Models</i>			
SegFormer (Original)	MiT-B0	0.535 ± 0.005	0.428 ± 0.029
SegFormer (FPN)	MiT-B2	0.637 ± 0.011	0.637 ± 0.015
U-Net (ResNet)	ResNet34	0.756 ± 0.016	0.605 ± 0.007
U-Net (Transformer)	MiT-B2	0.673 ± 0.009	0.682 ± 0.028

CWT



Dead



Resultados

ID	Dataset	Filename	Baseline	HumanExpert	SegPlantFormer	U-Net (ResNet)	U-Net (Transf)	SegFormer (FPN)
1	CWT	cwt3_6.jpg	0.619	0.675	0.682	0.885	0.870	0.399
2	CWT	cwt2_65.jpg	0.229	0.403	0.469	0.318	0.770	0.660
3	CWT	cwt1_43.jpg	0.571	0.637	0.755	0.874	0.966	0.868
4	CWT	cwt4_3.jpg	0.003	0.003	0.004	0.608	0.608	0.597
5	CWT	cwt2_23.jpg	0.374	0.420	0.577	0.873	0.731	0.850
6	CWT	cwt2_15.jpg	0.424	0.609	0.293	0.714	0.203	0.034
7	CWT	cwt3_75.jpg	0.498	0.546	0.542	0.706	0.548	0.457
8	CWT	cwt1_75.jpg	0.635	0.762	0.774	0.988	0.436	0.919
9	CWT	cwt2_76.jpg	0.254	0.279	0.385	0.788	0.688	0.788
10	CWT	cwt1_73.jpg	0.601	0.572	0.701	0.805	0.907	0.794
1	Dead	dead1_29.jpg	0.146	0.669	0.641	0.763	0.870	0.618
2	Dead	dead1_3.jpg	0.000	0.008	0.003	0.823	0.484	0.644
3	Dead	dead2_42.jpg	0.520	0.520	0.530	0.439	0.566	0.562
4	Dead	dead1_40.jpg	0.158	0.370	0.493	0.382	0.662	0.582
5	Dead	dead2_26.jpg	0.455	0.455	0.630	0.617	0.826	0.780

DEAD

CWT

Input images

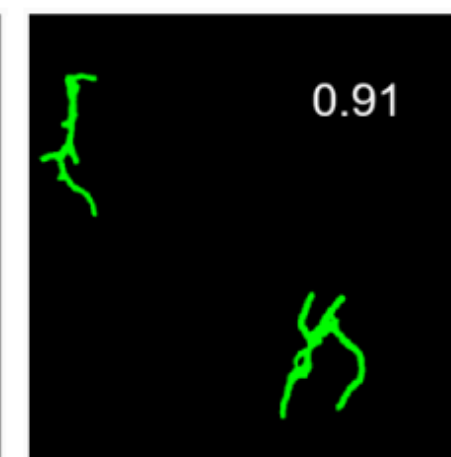
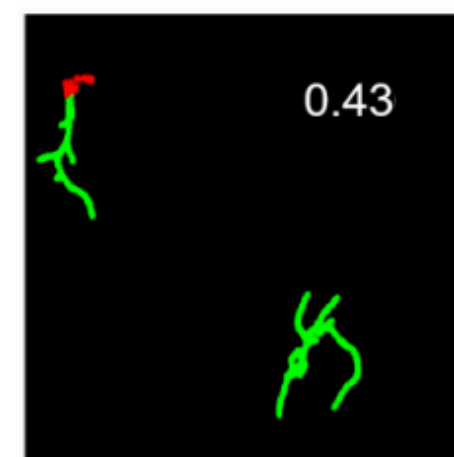
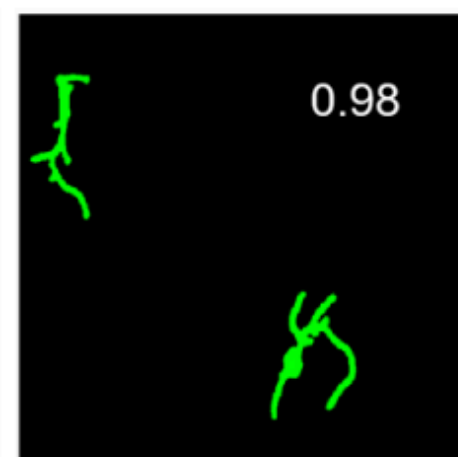
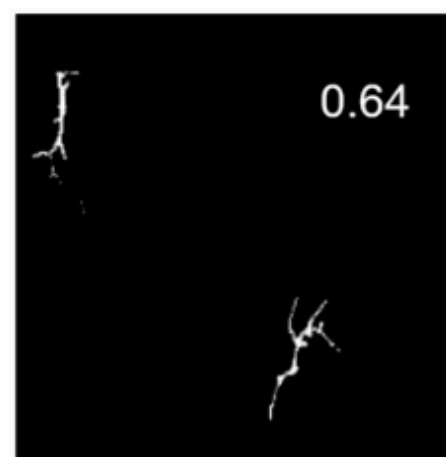
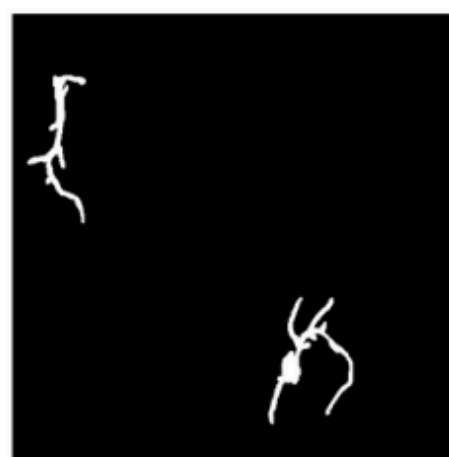
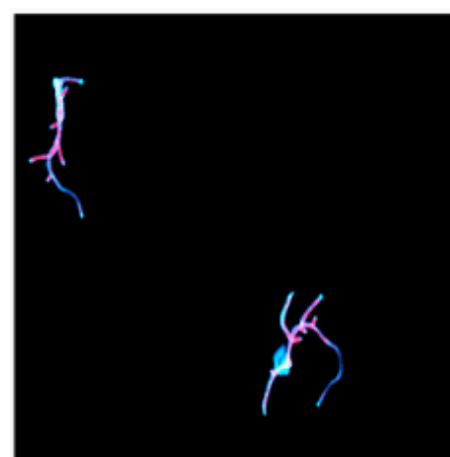
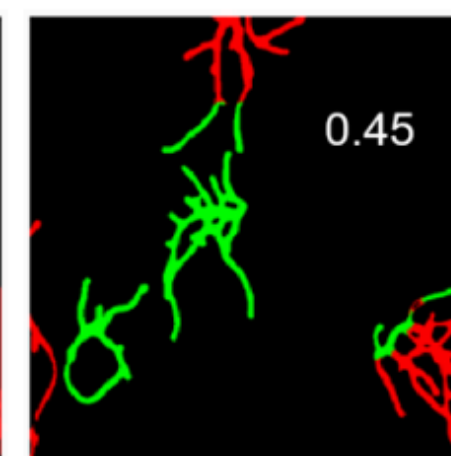
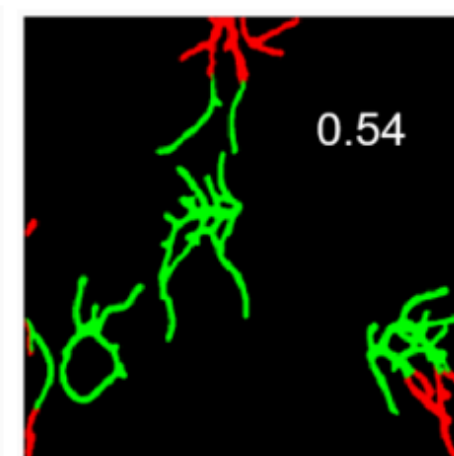
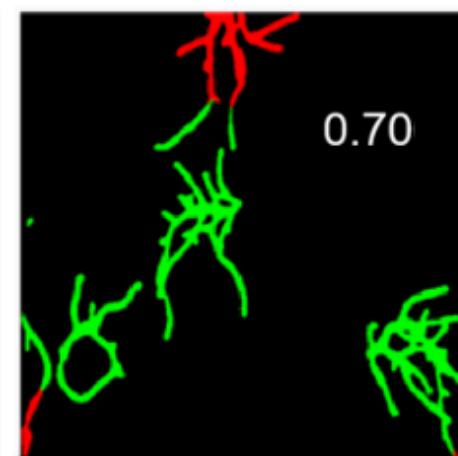
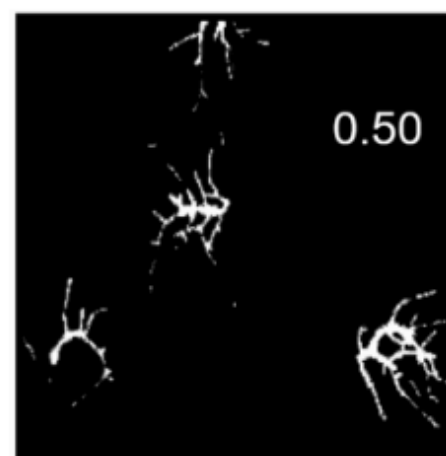
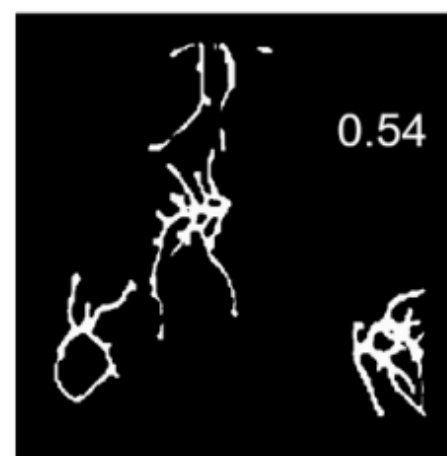
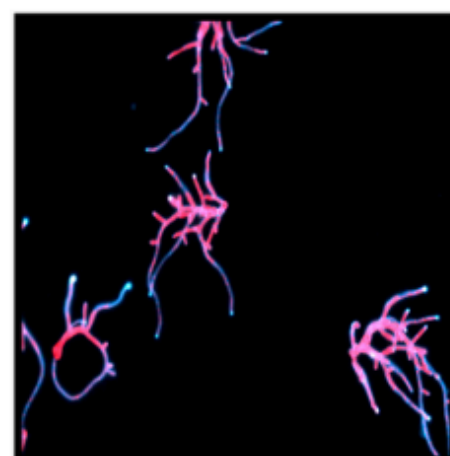
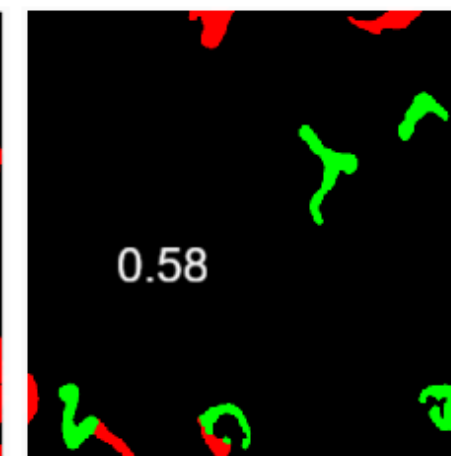
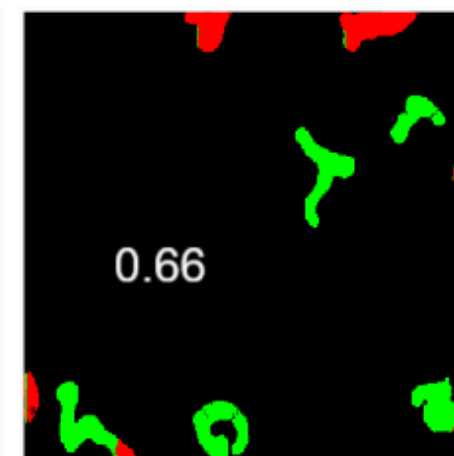
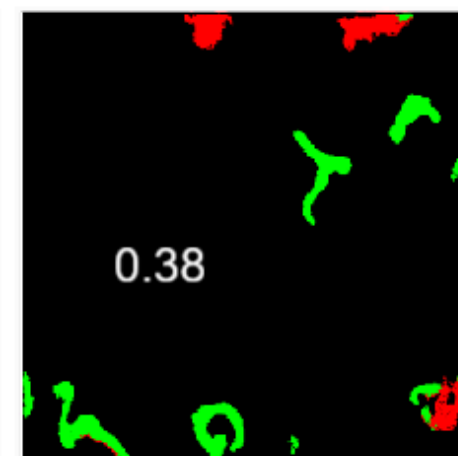
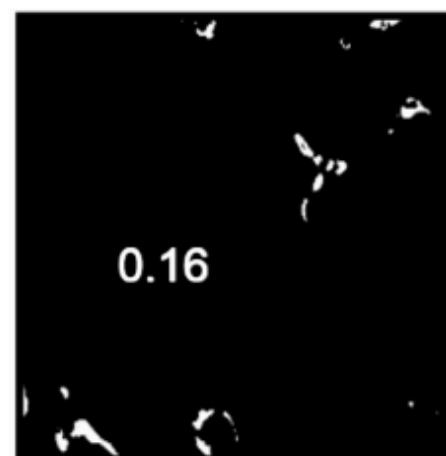
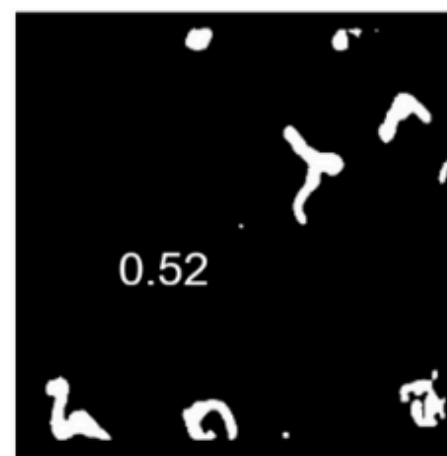
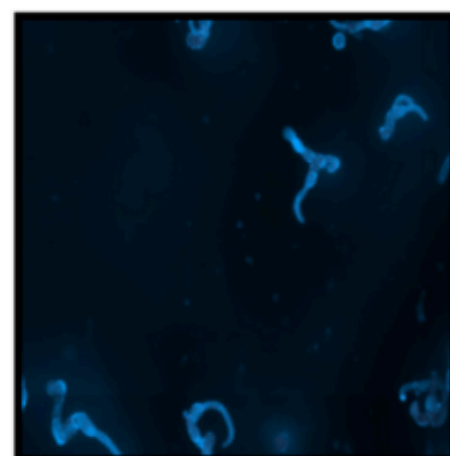
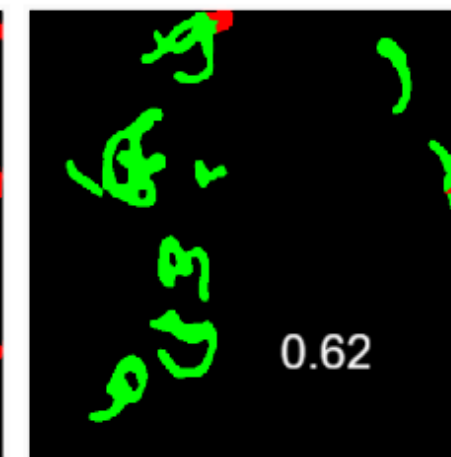
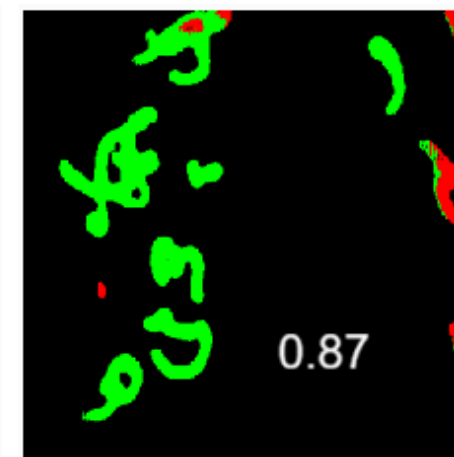
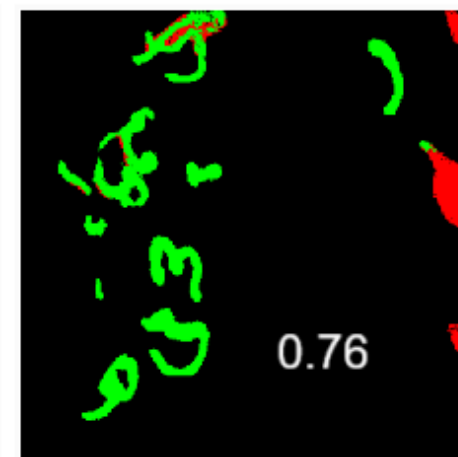
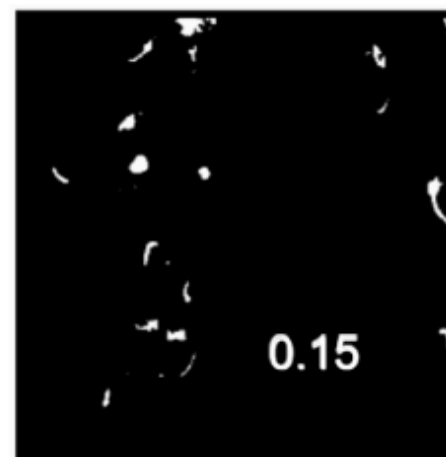
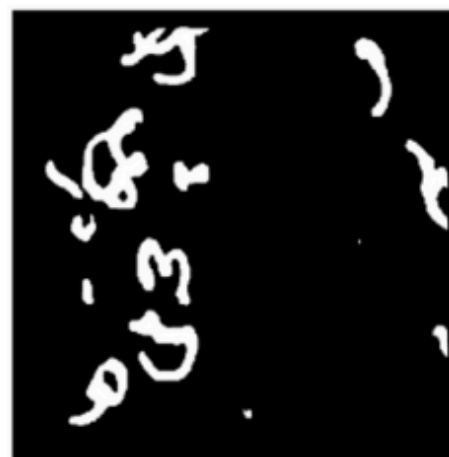
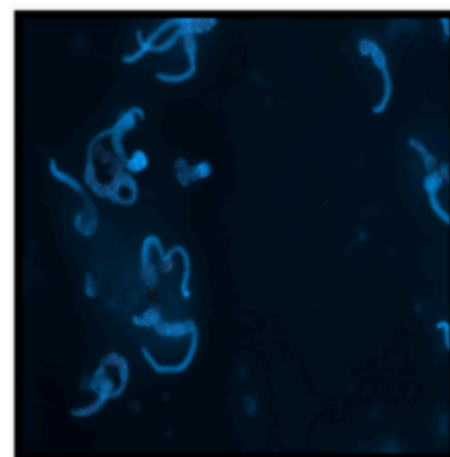
Ground truth masks

Predicted masks
PlantFormerPredicted masks
Traditional method

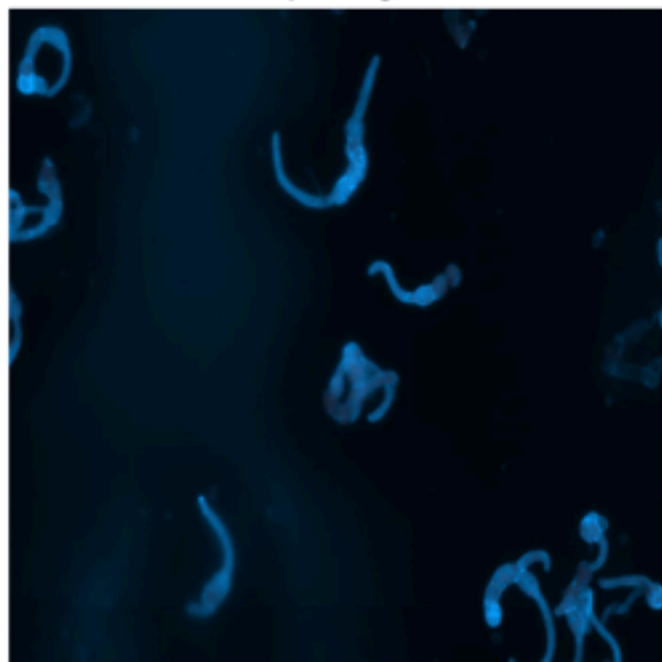
U-Net (ResNet)

U-Net
(Transformer)

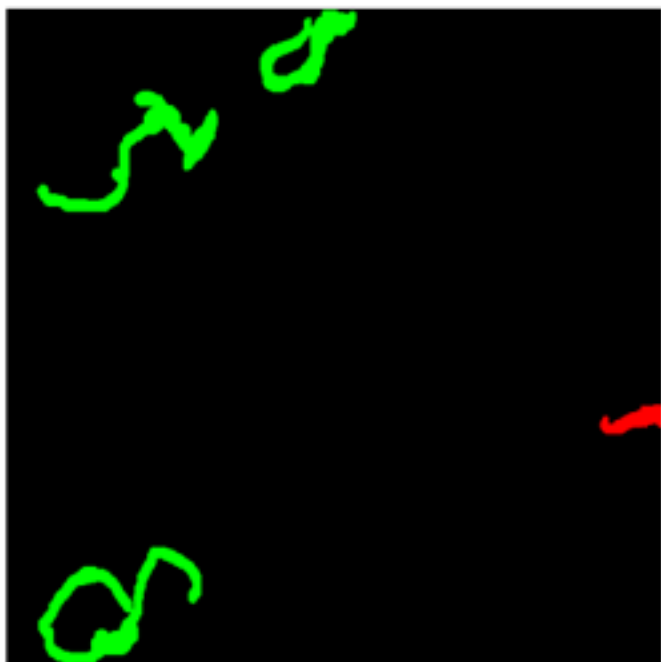
Segformer (FPN)



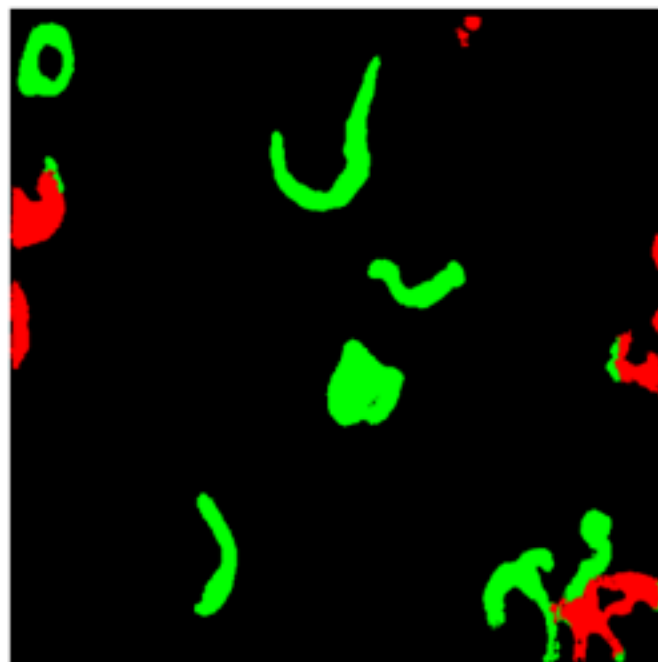
ID 2: dead1_3.jpg
Input Image



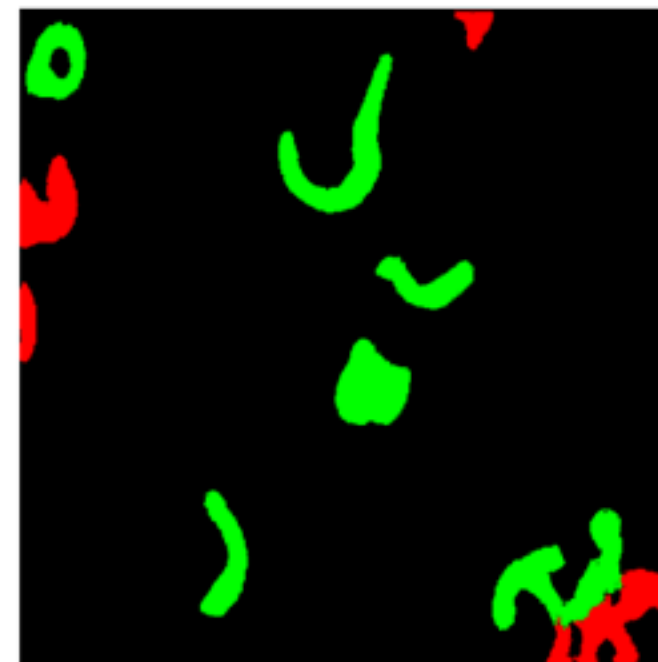
Ground Truth



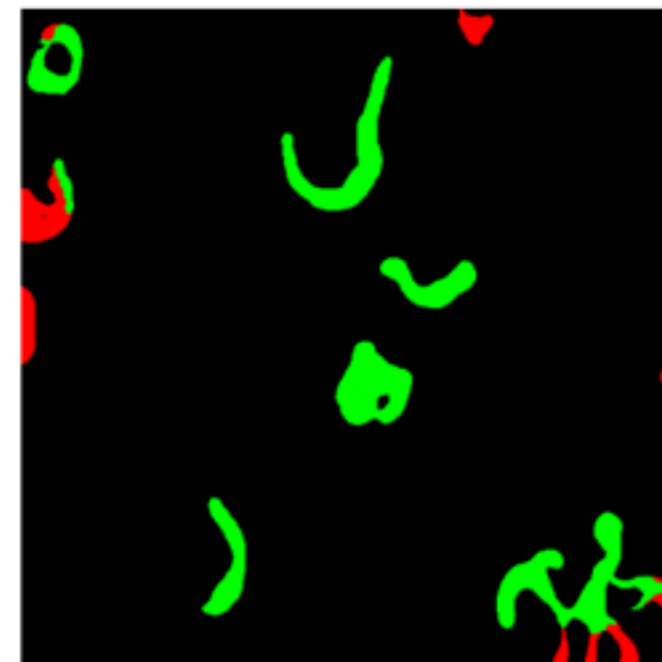
U-Net (ResNet)
IoU: 0.000

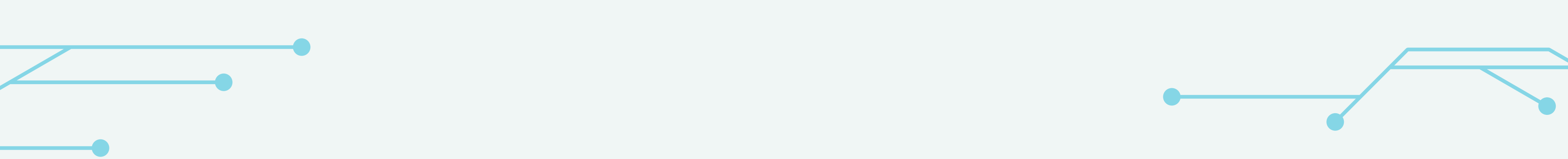


U-Net (Transformer)
IoU: 0.004



SegFormer (FPN)
IoU: 0.006





**MUCHAS
GRACIAS**